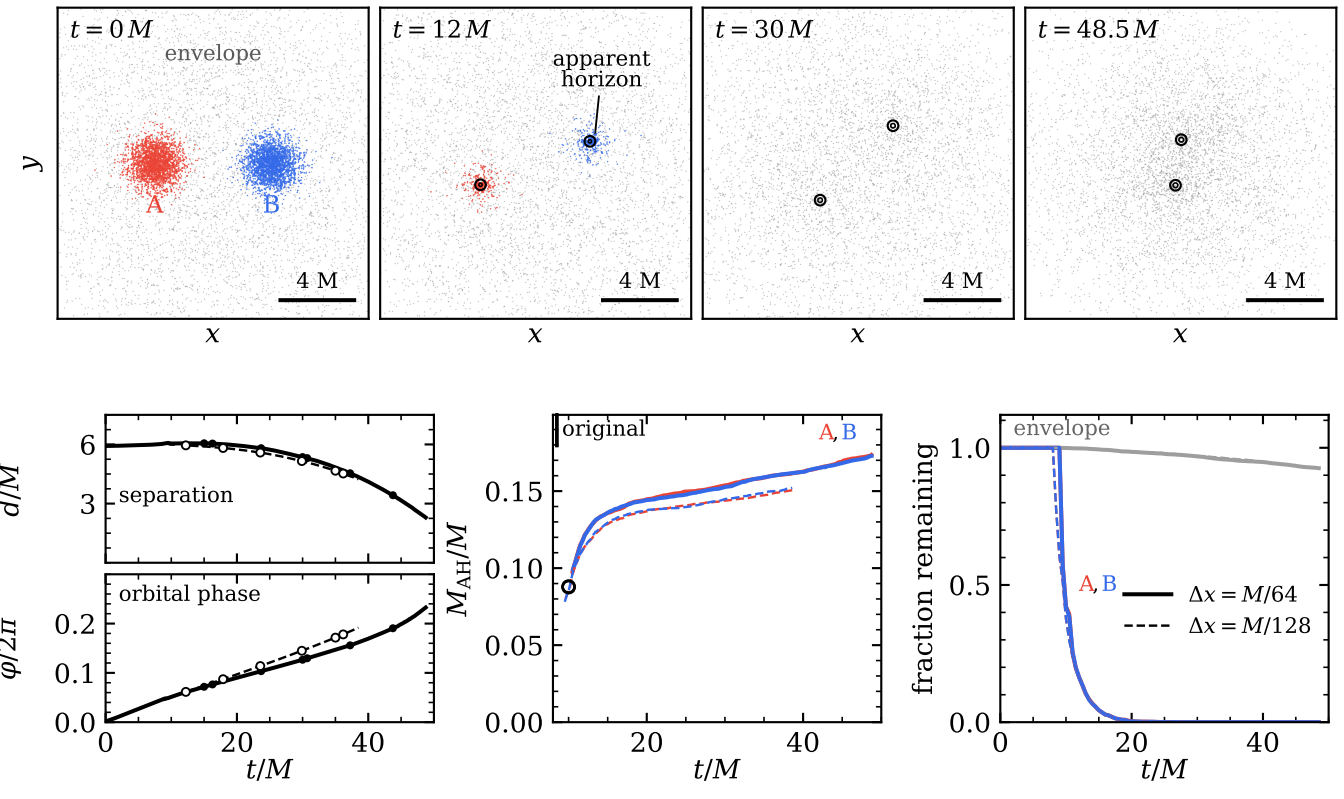

t = 0 M
envelope
A
B
4 M
t = 12 M
apparent
horizon
4 M
t = 30 M
4 M
t = 48.5 M
4 M
y
x
x
x
x
d/M
6
3
separation
orbital phase
φ/2π
0.2
0.1
0.0
0
20
40
t/M
original
A, B
M_AH/M
0.15
0.10
0.05
0.00
20
40
t/M
envelope
fraction remaining
1.0
0.5
0.0
A, B
Δx = M/64
Δx = M/128
0
20
40
t/M